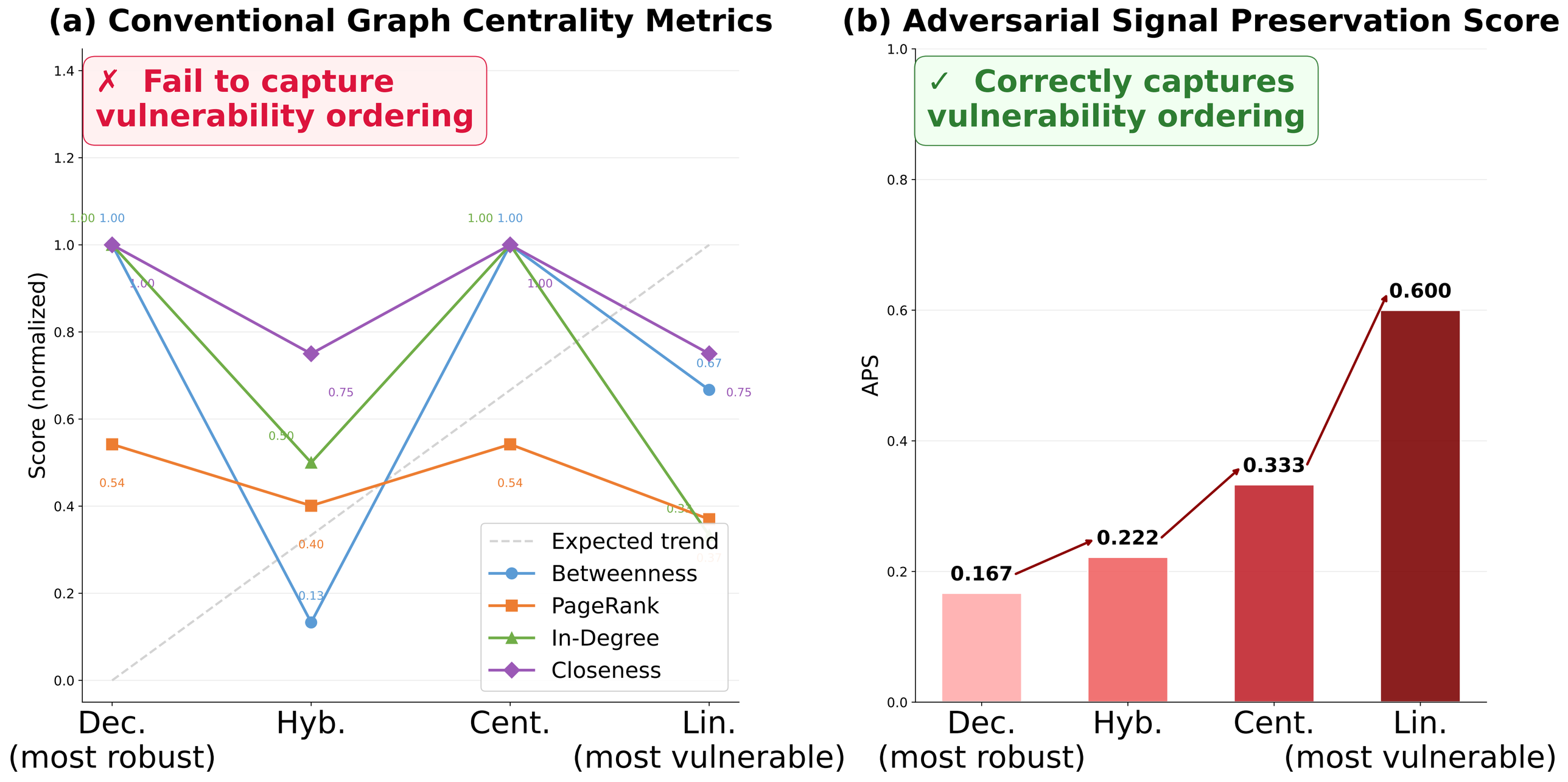
(a) Conventional Graph Centrality Metrics
✗ Fail to capture vulnerability ordering
Score (normalized)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
1.00 1.00
1.00
0.75
0.50
0.54
0.40
0.13
1.00 1.00
1.00
0.67
0.75
0.54
0.33
0.37
Expected trend
Betweenness
PageRank
In-Degree
Closeness
Dec.
(most robust)
Hyb.
Cent.
Lin.
(most vulnerable)
(b) Adversarial Signal Preservation Score
✓ Correctly captures vulnerability ordering
APS
1.0
0.8
0.6
0.4
0.2
0.0
0.167
0.222
0.333
0.600
Dec.
(most robust)
Hyb.
Cent.
Lin.
(most vulnerable)